

MEC-203

QUANTITATIVE METHODS



Volume-1

QUANTITATIVE METHODS

(Vol. 1)



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COURSE PREPARATION TEAM

Unit	Block/Unit Title	Unit Writer
BLOCK 1	REVIEW OF BASIC MATHEMATICS	
Unit 1	Fundamental Concepts of Mathematics	Dr. Meenakshi Sridhar, Associate Professor, Department of Computer Sciences, Rajdhani College, University of Delhi, Delhi
Unit 2	Overview of Basic Methods of Mathematics	
Unit 3	Relations and Functions	
Unit 4	Co-ordinate Geometry and Representation of Functions	Prof. Gopinath Pradhan
BLOCK 2	LINEAR ALGEBRA	
Unit 5	Matrix Algebra	Sunando Basu
Unit 6	Vector Analysis	
Unit 7	Vectors and Matrices	
Unit 8	Vector and Matrix Representations of Linear Equations	Dr. Meenakshi Sridhar, Associate Professor, Department of Computer Sciences, Rajdhani College, University of Delhi, Delhi
BLOCK 3	CALCULUS	
Unit 9	Limit and Continuity	Sunando Basu & Prof. Gopinath Pradhan
Unit 10	Differential Calculus: Functions of One Variable	
Unit 11	Differential Calculus: Functions of Several Variables	
Unit 12	Integration: Introduction and Techniques	Prof. Gopinath Pradhan
BLOCK 4	REAL ANALYSIS	
Unit 13	Real Analysis	Dr. Meenakshi Sridhar, Associate Professor, Department of Computer Sciences, Rajdhani College, University of Delhi, Delhi
Unit 14	Calculus of Several Variables	
Unit 15	Basic Concepts of Metric Space and Point Set Topology	
BLOCK 5	EXTREME VALUES AND OPTIMISATION	
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Unit 17	Unconstrained and Constrained Optimisation	Rittwik Chatterjee
Unit 18	Advanced Topics in Optimisation– I	Dr. Ram Prasad Yadav, Assistant Professor, Department of Mathematics, SRM University, Sonepat, Haryana
Unit 19	Advanced Topics in Optimisation – II	

BLOCK 6	ECONOMIC DYNAMICS	
Unit 20	Difference Equations: Theory and Economic Applications	Prof. Gopinath Pradhan
Unit 21	Differential Equations	
BLOCK 7	DYNAMIC OPTIMISATION	
Unit 22	Intertemporal Optimisation-I	Mr. Debasish Dash, Independent Researcher New Delhi
Unit 23	Intertemporal Optimisation-II	
Unit 24	Economic Applications of Dynamic Optimisation	
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Unit 26	Probability Distributions-I	
Unit 27	Probability Distributions-II	
BLOCK 9	INFERENTIAL STATISTICS	
Unit 28	Sampling Theory	Biswadeep Basu
Unit 29	Sampling Distributions	
Unit 30	Estimation	Biswadeep Basu (updatation by Ms. Nivedita Mullick)
Unit 31	Hypothesis Testing	Biswadeep Basu

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COURSE INTRODUCTION

The importance of quantitative methods in economic analysis cannot be undermined. Apart from helping us in quantitative expression of attributes, it allows greater precision and accuracy in measurement. We observe an ever-increasing usage of mathematics and statistics in all branches of economics. This calls for an extensive and compulsory study of quantitative methods in the present academic programme. After going through this course a student will be able to:

- use mathematics as a language to represent economic ideas, concepts, phenomena, and relationship among variables;
- apply mathematics as a tool to explain various economic phenomena, events and situations at local, regional, national and global level;
- acquire basic knowledge of mathematical methods and statistical tools to explain economic theory;
- apply optimization techniques to find out optimum level of objective functions pursued by economic agents; and
- apply statistical tools for data presentation, data analysis, hypothesis formulation, and testing of hypothesis.

The course is divided into **nine blocks** comprising **thirty-one Units**.

Block 1, entitled **Review of Basic Mathematics**, provides an overview of fundamental concepts of mathematics. It comprises four Units. Unit 1 deals with the basic concepts of set theory while Unit 2 discusses algebraic concepts and expressions such as polynomial functions. Unit 3 deals with relations and functions and their types. Representation of such functions through coordinate geometry is the subject matter of Unit 4.

Block 2, entitled **Linear Algebra**, comprises four Units. Unit 5 provides an overview of matrix algebra, particularly the concepts of determinants, cofactors and matrix operations. Unit 6 deals with vector space and different types of vector operations. Unit 7 moves into further issues related to vectors and matrices such as characteristic values, Eigen values and vector differentiation. Unit 8 deals with linear independence of vectors and certain concepts such as R^n space.

Block 3, comprising four Units, provides certain introductory ideas on **Calculus**. Unit 9 focuses on limit and continuity of a function. Unit 10 introduces the concept of differentiation of a function, particularly the one-variable case. In the process it interprets the concept and provides several illustrations from economics. Unit 11 continues with differentiation of functions in several-variables case. Unit 12 deals with the concept and applications of integration in the field of economics.

Block 4, entitled **Real Analysis**, comprises three Units. Unit 13 introduces the students to n -dimensional real space and mathematical logic. Unit 14 extends the ideas discussed in the previous Unit further to the several-variables case. Unit 15 discusses basic concepts

of matrix space and point set topology. This block includes several important topics such as extreme value theorem, mean value theorem and Taylor's theorem.

Block 5, entitled **Extreme Values and Optimisation**, builds upon the previous two blocks. It comprises four Units. Unit 16 deals with the concept of optimization in one-variable and several-variables cases. Through several examples, the Unit illustrates how maxima and minima of a function can be identified. Unit 17 extends the topic further to the constrained-optimisation scenario. Units 18 and 19 carry forward the topic of constrained optimisation to an advanced level and deal with several variants of constraints that find applications in the field of economics.

The title of **Block 6** is **Economic Dynamics**, and it deals with the time path followed by economic variables. Unit 20 deals with the theory and applications of difference equations, which treats time as a discrete variable. The solution to first order and second order difference equations are presented in the Unit. Unit 21 presents the continuous variables case and discusses the theory and applications of differential equations in the field of economics.

Block 7, titled **Dynamic Optimisation**, consists of three Units. Unit 22 introduces the students to the issue of intertemporal optimisation, that is, optimisation of variables in cases involving several time periods. It deals with the Euler's equation, its solution and applications in economics. Unit 23 continues with the topic of intertemporal optimisation and deals with on optimal control problems. Unit 24 focuses on economic applications of dynamic optimisation problems.

Block 8, entitled **Probability and Probability Distributions**, comprises three Units. Unit 25 deals with the concept of probability and various theorems on probability. For better elucidation the Unit provides a number of examples where various probability laws are applied. Unit 26 presents discrete probability distributions such as binomial and Poisson. Unit 27, in a similar vein, presents continuous probability distributions such as normal distribution.

Block 9, consisting of four Units, focuses on **Inferential Statistics**. Unit 28 discusses issues related to drawing of various types of samples from a population. The concept, properties and applications of sampling distribution are presented in Unit 29. In Unit 30 the issues related to estimation of parameters on the basis of a sample is discussed. The concluding unit of the course, Unit 31, deals with the formulation and testing of hypothesis and drawing inferences on the basis hypothesis testing.

MATHEMATICAL SYMBOLS/ NOTATIONS USED IN THE COURSE

Symbol	Symbol Name/Meaning Attached
$[a,b]$	Closed interval
(a,b)	Open interval
$\in$	Element of
Δ	Delta/Change in
dx	Differential
$\lim_{\Delta x \rightarrow 0} f(x)$	Limit
$\ln$	Natural log
$\int_a^b f(x)dx$	Definite Integral
$\forall$	Universal quantifier: "for all"
$\exists$	Existential quantifier: "there exists"
$\exists_1$	Existential quantifier: "there exists exactly one"
$\exists_{>1}$	Existential quantifier: "there exists more than one"
$\neg$	Negation; $\neg P$ "not P"
$\wedge$	Conjunction: "and"
$\vee$	Disjunction: "or"
$\Rightarrow$	Implication; $p \Rightarrow q$ "if p then q"
$\Leftrightarrow$	Equivalence: "if and only if"
$a, x \in \mathbb{R}$	Numbers, scalars
$a = (a_1, a_2, \dots, a_n) \in \mathbb{R}_+^n$	Vector of parameters
$x = (x_1, x_2, \dots, x_n) \in \mathbb{R}_+^n$	Vector of variables
A, X	Sets
A, X	Matrices (with dimensions m by n) with m rows and n columns
$x \sim y$	Vector x is indifferent with respect to vector y
$\succ$	Strong preference relation
$x \succ y$	Vector x is strongly preferred over vector y
$\succsim$	(weak) Preference relation
$x \succsim y$	Vector x is (weakly) preferred over vector y
$\mathbb{R}$	Set of real numbers
$\mathbb{R}_+$	Set of nonnegative real numbers
$\text{int}\mathbb{R}_+$	Set of positive real numbers
$\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}$	n – Dimensional space of real numbers, Cartesian product of set $\mathbb{R}$
$\mathbb{R}_+^n = \{x \in \mathbb{R}^n \mid x \geq 0\} \subset \mathbb{R}^n$	Nonnegative orthant (subspace) of set $\mathbb{R}^n$
$\text{int}\mathbb{R}_+^n \subset \mathbb{R}_+^n$	Interior of set $\mathbb{R}^n$
$f : X \rightarrow Y$	Function, X – Domain of a function (set of arguments of a function) Y – Codomain of a function (set of values of a function)